

Assembly Procedure

Table 1

Required Tools			
Tool	Part Number	Part Description	Qty
A	FT-1411	Lifting Bracket	2
B	138-7575	Link Bracket	2
C	1P-0510	Driver Group	1
D	138-7574	Link Bracket	2
F	-	Loctite 243	-

Note: Check all of the O-ring seals and the components for wear or for damage. Replace the components, if necessary. Lubricate all of the O-ring seals lightly with the lubricant that is being sealed. Clean all of the components with a cloth that is free of loose material (lint).

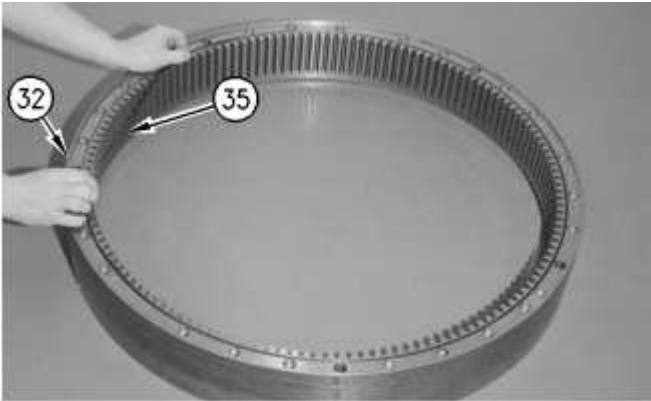


Illustration 1 g00648788

1. Install O-ring seal (35) on ring (32). Turn over ring (32).

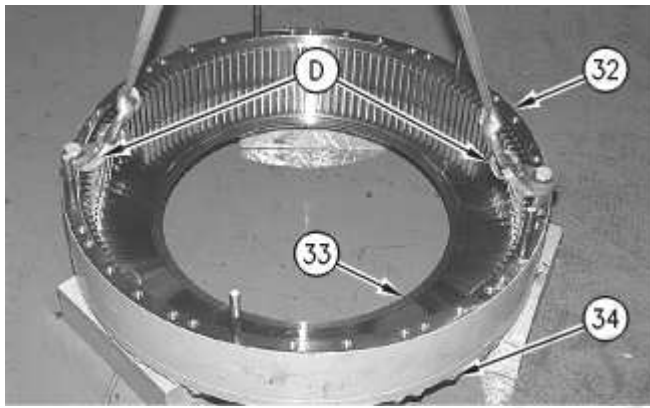


Illustration 2

g00648777

2. Attach Tooling (D) and a suitable lifting device to ring (32). Position ring (32) on plate (33). The weight of ring (32) is approximately 100 kg (220 lb).
3. Install bolts (34) and the hard washers that hold plate (33) to ring (32). Tighten bolts (34) to a torque of $310 \pm 25 \text{ N}\cdot\text{m}$ ($229 \pm 18 \text{ lb}\cdot\text{ft}$).

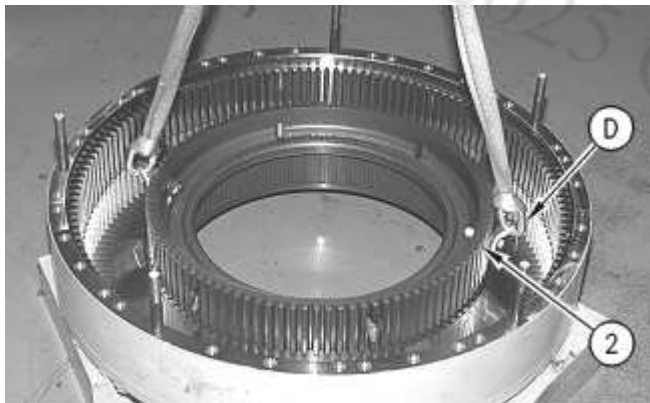


Illustration 3

g00648789

4. Attach Tooling (D) and a suitable lifting device to brake hub (2). Position brake hub (2) on plate (33), as shown. The weight of brake hub (2) is approximately 45 kg (100 lb).

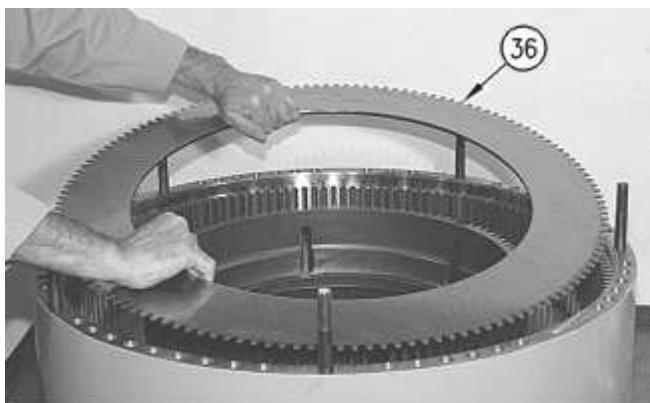


Illustration 4

g00648790

5. Install bottom damper assembly (36) with the face toward the plate.

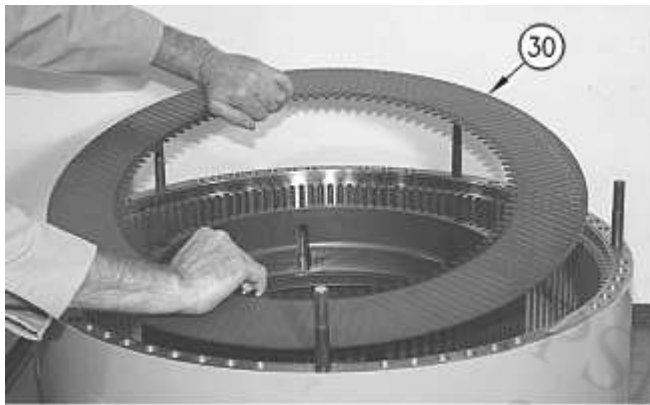


Illustration 5

g00648791

6. Lubricate each friction disc (30) with a light coat of the oil that is used in the service and parking brake. Install friction disc (30).



Illustration 6

g00648775

7. Install outer plates (31) and the remaining friction discs (30) in alternating order.



Illustration 7

g00648774

8. Install top damper assembly (29) with the face upward, as shown.

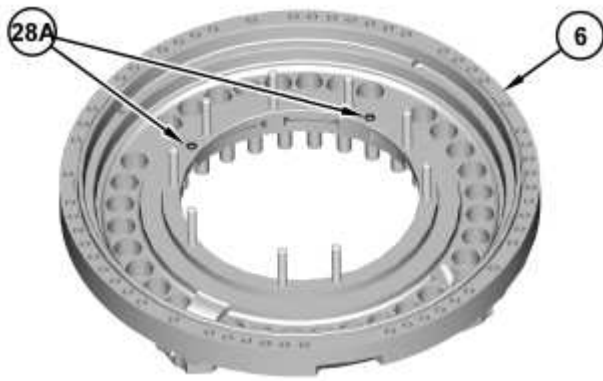


Illustration 8

g03861180

9. Install two seals (28A) in anchor assembly (6).

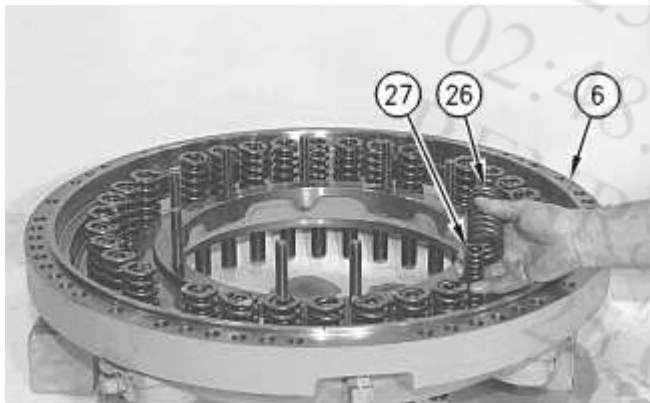


Illustration 9

g00648793

10. Install outer springs (26) and install inner springs (27) in anchor (6).

WARNING

Improper assembly of parts that are spring loaded can cause bodily injury.

To prevent possible injury, follow the established assembly procedure and wear protective equipment.

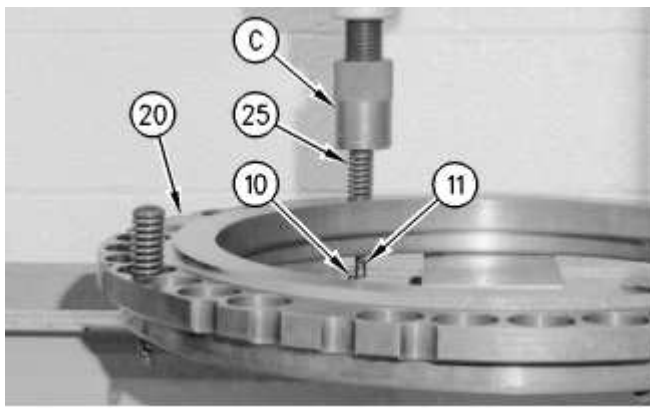


Illustration 10

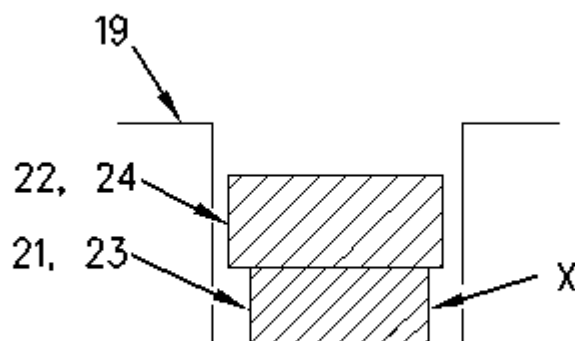
g00648767

11. Position piston (20) and Tooling (C) in a suitable press, as shown. Install spring guide (11) and spring (25) in piston (20). Use the press and Tooling (C) to compress spring (25) until pin (10) can be installed in the second hole of spring guide (11). Slowly release the pressure on spring (25).
12. Repeat Step 11 for the remaining three springs (25), spring guides (11), and pins (10).



Illustration 11

g00648766



View of an incorrectly installed seal (21, 23)

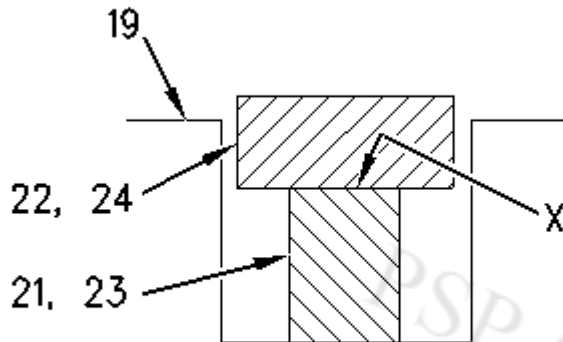


Illustration 13

g00713644

View of a correctly installed seal (21, 23)

Note: Be sure that yellow side (X) of inside seal (21, 23) is facing outward.

13. Install seal (21) and ring seal (22) on housing (19). Install seal (23) and ring seal (24) on housing (19).

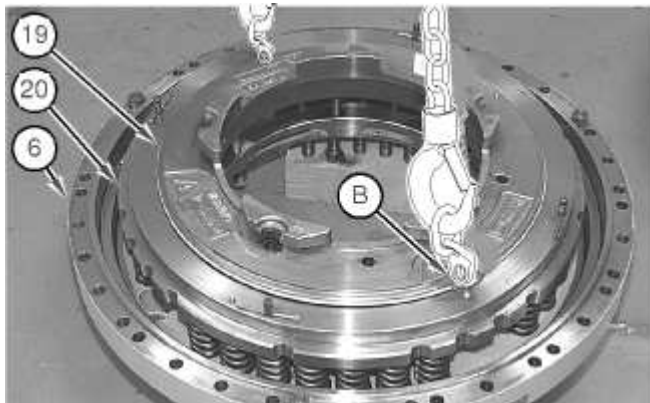


Illustration 14

g01033968

14. Use two people to install housing (19) in piston (20). Do not seat the housing at this time.
15. Install Tooling (B) and a suitable lifting device in piston (20). Lift housing (19) and piston (20) as an assembly into position on anchor (6), as shown. The weight of the assembly is approximately 70 kg (154 lb). Position the assembly so that the four inner springs (25) (not shown) are aligned with the correct outer springs in anchor (6).
16. Align the holes in housing (19) with the nine studs by rotating the housing.

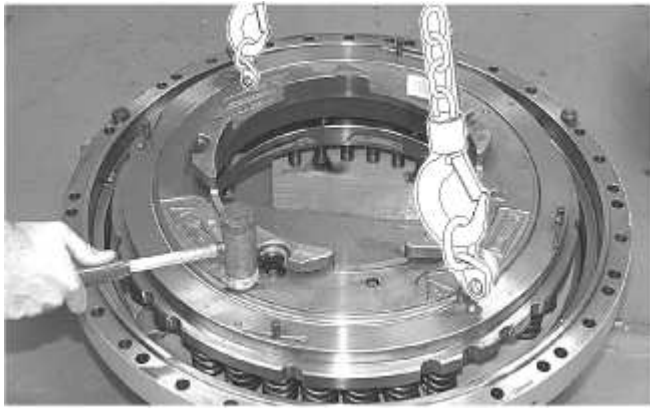


Illustration 15

g01033982

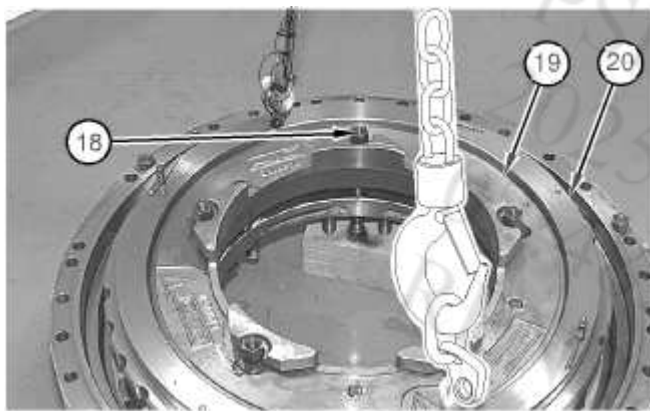


Illustration 16

g01033984

17. Use a soft faced hammer to push housing (19) downward evenly over the studs until nuts (18) can be started.

 WARNING

Improper assembly of parts that are spring loaded can cause bodily injury.

To prevent possible injury, follow the established assembly procedure and wear protective equipment.

-
18. Install five nuts (18) that hold housing (19) to piston (20). Slowly tighten each nut (18) by one revolution at a time to pull housing (19) and piston (20) downward evenly. After all five nuts (18) have been installed, tighten the nuts to a torque of 300 ± 15 N·m (221 ± 11 lb ft) in a crisscross pattern.
-

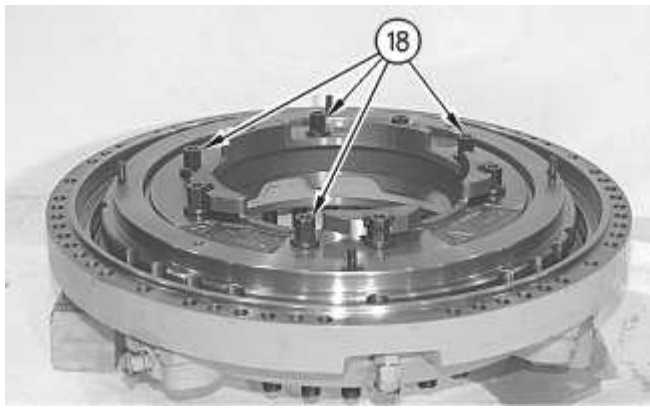


Illustration 17

g00648760

19. Install the four remaining bolts (18). Tighten the bolts to a torque of $300 \pm 15 \text{ N}\cdot\text{m}$ ($221 \pm 11 \text{ lb ft}$) in a crisscross pattern.

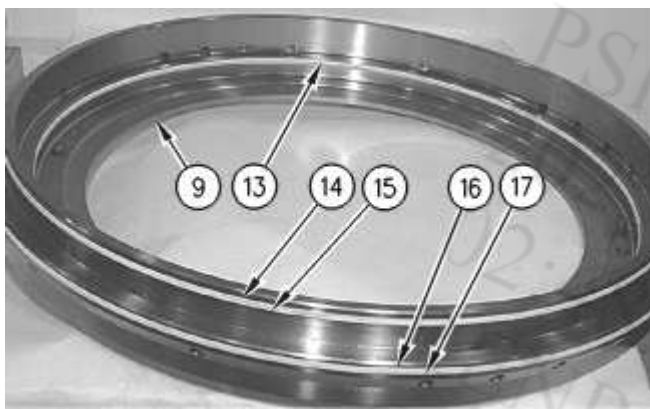


Illustration 18

g00648759

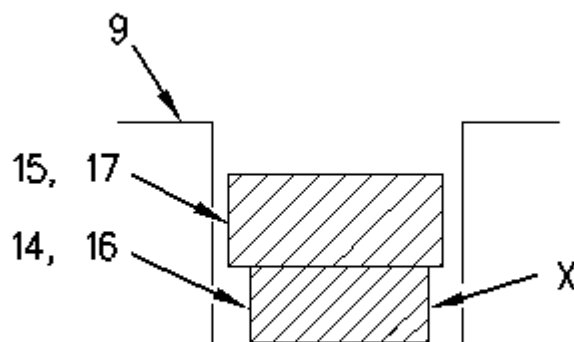


Illustration 19

g00713664

View of an incorrectly installed seal (14, 16)

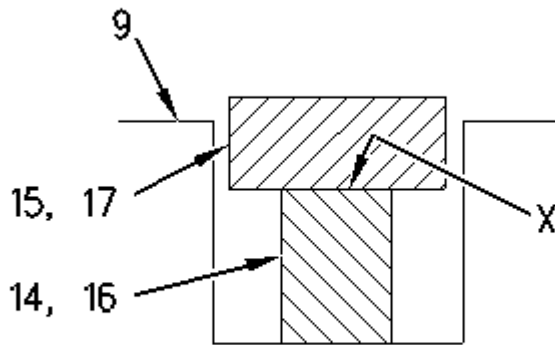


Illustration 20

g00713667

View of a correctly installed seal (14, 16)

20. Install seal ring (13) on piston (9).

21. Install seal (14) and ring seal (15) on piston (9).

Note: Be sure that yellow side (X) of seal (14) is facing outward.

22. Install ring seal (16) and ring seal (17) on piston (9). Use two people to turn over the piston.

Note: Be sure that yellow side (X) of ring seal (16) is facing outward.



Illustration 21

g01033778

23. Install Tooling (B) and a suitable lifting device in piston (9). Lift piston (9) into position on anchor assembly (6). The weight of piston (9) is approximately 45 kg (100 lb).



WARNING

Improper assembly of parts that are spring loaded can cause bodily injury.

To prevent possible injury, follow the established assembly procedure and wear protective equipment.

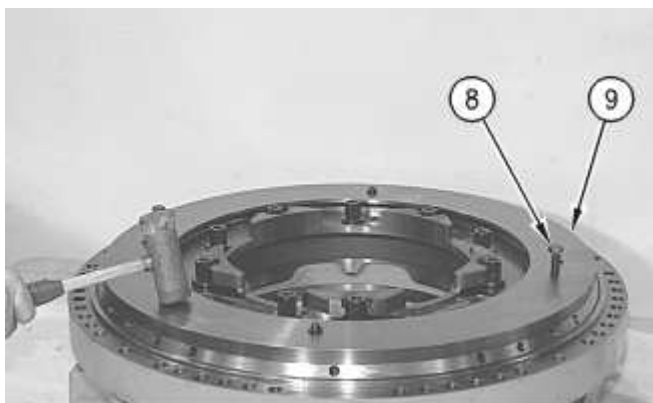


Illustration 22

g00648753

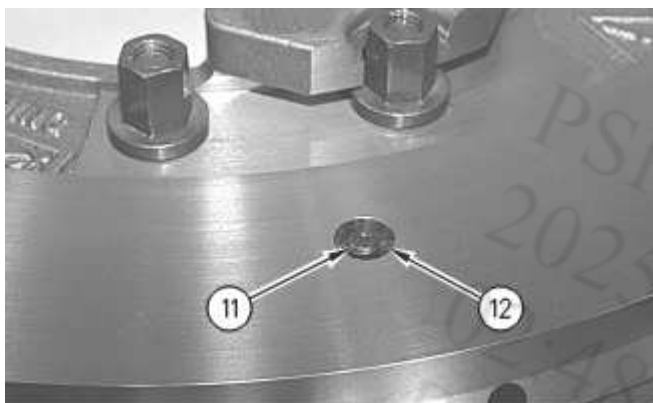


Illustration 23

g00648754

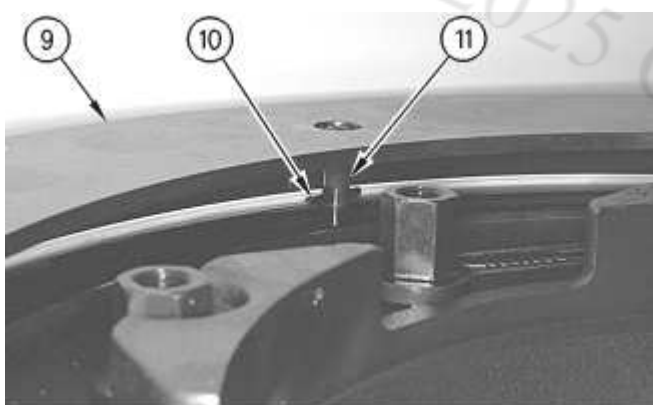


Illustration 24

g00648750

24. Use a soft faced hammer to push piston (9) downward until four pins (12) can be installed in four spring guides (11).
25. Install two 1/2 in - 13 NC forcing screws (8) in piston (9), as shown. Tighten the forcing screws evenly until there is approximately 25 mm (1 inch) between the two pistons.

26. Remove four pins (10) from the holes of four spring guides (11) beneath piston (9).
27. Loosen two forcing screws (8) evenly until piston (9) is correctly positioned on the anchor assembly. Remove the forcing screws.

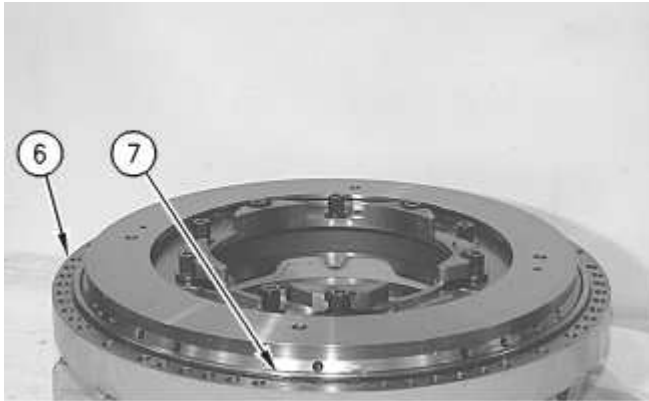


Illustration 25

g00648747

28. Install O-ring seal (7) on anchor assembly (6).

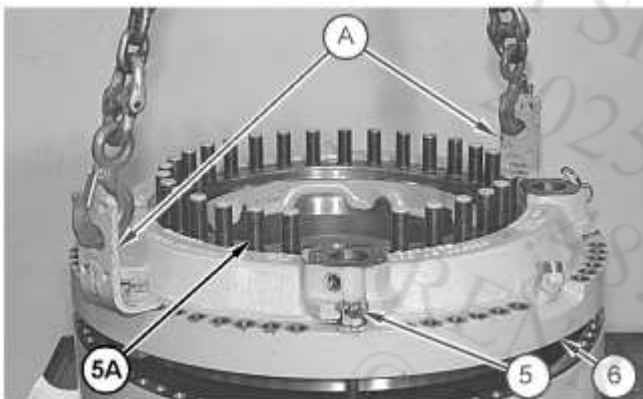


Illustration 26

g06620164

29. Turn over anchor assembly (6). Attach Tooling (A) and a suitable lifting device to anchor assembly (6). Lift anchor assembly (6) into position on the ring, as shown. The weight of the anchor assembly is approximately 259 kg (571 lb).
 30. Install five nuts (5) that hold anchor assembly (6) in place. Slowly tighten each nut (5) by one revolution at a time to pull anchor assembly (6) downward evenly. After all five nuts (5) have been installed, tighten the nuts to a torque of $300 \pm 15 \text{ N}\cdot\text{m}$ ($221 \pm 11 \text{ lb ft}$) in a crisscross pattern.
 31. If studs (5A) are removed, install new studs with the threadlock patch into the brake anchor or apply Tooling (F) to the threads of studs (5A) and install the studs. Tighten studs (5A) to a torque of $400 \pm 60 \text{ N}\cdot\text{m}$ ($295 \pm 44 \text{ lb ft}$).
-

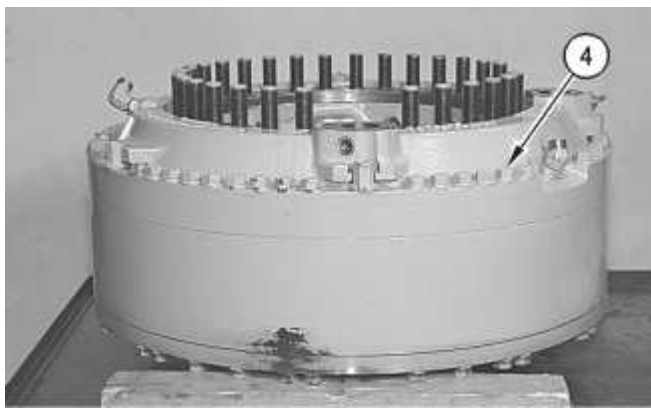


Illustration 27

g00648737

32. Install bolts (4) and the hard washers that hold the anchor assembly to the ring. Tighten the bolts to a torque of 310 ± 25 N·m (229 ± 18 lb ft).



Illustration 28

g00648734

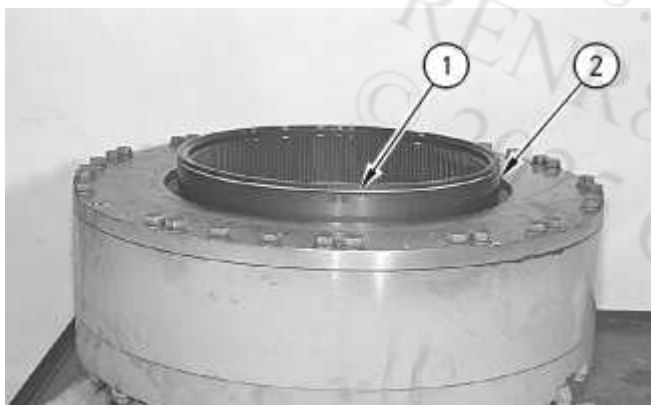


Illustration 29

g00648729

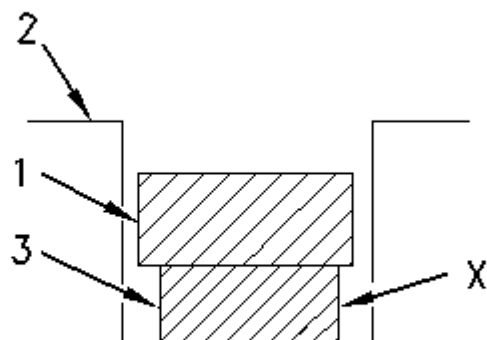


Illustration 30
View of an incorrectly installed seal ring (3)

g00713673

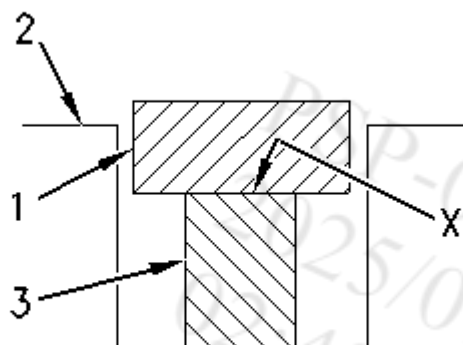


Illustration 31
View of a correctly installed seal ring (3)

g00713674

33. Turn over the rear service and parking brake, as shown. The weight of the service and parking brake is 630 kg (1390 lb). Install seal ring (3) on brake hub (2).

Note: Be sure that yellow side (X) of seal ring (3) is facing outward.

34. Install ring seal (1) on brake hub (2).

End By:

- a. Install the rear service and parking brake. Refer to Disassembly and Assembly, "Service and Parking Brake - Install".

PSP-0007BF75

2025/01/17

02:48:26+07:00

i01317345

© 2025 Caterpillar Inc.